

Problem 1. Lines and Circles. The parametrized curve in part (a) is a line. The parametrized curve in part (b) is a circle. In each case, compute the velocity and speed at time t . Also eliminate t to find an equation for the curve in terms of x and y .

- (a) $(x, y) = (p + ut, q + vt)$ where p, q, u, v are constants.
- (b) $(x, y) = (a + r \cos(\omega t), b + r \sin(\omega t))$ where a, b, r, ω are constants.

Problem 2. Semi-Cubical Parabola. Consider the parametrized curve

$$(x, y) = (t^2, t^3).$$

- (a) Eliminate t to find an equation relating x and y . [Hint: Note that $y/x = t$.]
- (b) Compute the velocity and speed at time t .
- (c) Find the slope of the tangent line at time t .
- (d) Use the information in (b) and (c) to sketch the curve for t from $-\infty$ to ∞ .

Problem 3. The Cycloid. The cycloid is an interesting curve whose arc length can be computed by hand. It is parametrized by

$$(x, y) = (t - \sin t, 1 - \cos t).$$

- (a) Check that the slope of the tangent at time t is $\sin t / (1 - \cos t)$. Use this information to sketch the curve between $t = 0$ and $t = 2\pi$. [Hint: The slope goes to infinity when $t \rightarrow 0$ from the right and when $t \rightarrow 2\pi$ from the left. You do not need to prove this.]
- (b) Compute the arc length between $t = 0$ and $t = 2\pi$. [Hint: You will need the trig identities $\sin^2 t + \cos^2 t = 1$ and $1 - \cos t = 2 \sin^2(t/2)$.]

Problem 4. A Triangle in the Plane. Consider the following points in $\mathbb{R}^2$:

$$P = (1, 3), \quad Q = (-1, 2), \quad R = (2, 4).$$

- (a) Draw the three points together with the midpoints $(P + Q)/2$, $(P + R)/2$, $(Q + R)/2$ and the center of mass $(P + Q + R)/3$.
- (b) Find the coordinates of the three side vectors $\mathbf{u} = \vec{PQ}$, $\mathbf{v} = \vec{QR}$, $\mathbf{w} = \vec{PR}$.
- (c) Use the length formula to compute the three side lengths $\|\mathbf{u}\|$, $\|\mathbf{v}\|$, $\|\mathbf{w}\|$.
- (d) Use the dot product to compute the three angles of the triangle.

Problem 5. A Triangle in Space. Consider the following points in $\mathbb{R}^3$:

$$P = (1, 0, 0), \quad Q = (1, 1, 0), \quad R = (1, 1, 1).$$

- (a) Find the coordinates of the three side vectors $\mathbf{u} = \vec{PQ}$, $\mathbf{v} = \vec{QR}$, $\mathbf{w} = \vec{PR}$.
- (b) Use the length formula to compute the three side lengths $\|\mathbf{u}\|$, $\|\mathbf{v}\|$, $\|\mathbf{w}\|$.
- (c) Use the dot product to compute the three angles of the triangle.

Problem 6. Some Vector Arithmetic. Let $\mathbf{u}$ and $\mathbf{v}$ be any two vectors in 100-dimensional space. Use the properties of the dot product to show that

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2(\mathbf{u} \bullet \mathbf{v}).$$

[Hint: Start with the definition $\|\mathbf{u} - \mathbf{v}\|^2 = (\mathbf{u} - \mathbf{v}) \bullet (\mathbf{u} - \mathbf{v})$, then expand using FOIL.]